

IMRAN NAWAR

AI/ML Engineer

Peshawar, Pakistan

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EDUCATION

Bachelor of Science in Computer Science

Islamia College Peshawar

Sept 2020 - July 2024

Peshawar, Pakistan

- CGPA: 3.78/4.00
- Advisor: [Dr. Muhammad Sajjad](#), PI Digital Image Processing Lab
- Thesis: Deep Learning Assisted Autonomous Navigation over Economical Hardware

EXPERIENCE

Digital Image Processing (DIP) Lab

Research Assistant

July 2024 - Aug 2026

Peshawar, Pakistan

- Co-authored “NeuroShield-IoE: a cognitive, quantum-resilient and self-evolving cybersecurity framework for the Internet of Energy”, published in IEEE Network (*Impact Factor: 6.3, Q1*).
- Co-authored a second paper on “Democratizing autonomous driving research via economical hardware”, currently under review at IJVAS.
- Contributed to a UKRI-EPSCRC NIA proposal on AI risk modelling and differential privacy for healthcare systems (APP86824), with [Dr. Maqsood Mahmud](#) (Ulster University), currently under review.
- Worked as Teaching Assistant for five courses: Data Visualization (DSC 635) [G](#), Digital Image Processing (COMP416), Information Retrieval (COMP423), Artificial Intelligence (COMP315) [G](#) and Advanced Machine Learning (CSC 731). [\[Evaluation Material\]](#)

Undergraduate Student Researcher

Nov 2023 - June 2024

- Designed and built an autonomous navigation prototype on Jetson Nano edge hardware for my BS thesis.
- Developed an RFID-based door lock system using an Arduino microcontroller.

Corvit Networks

Artificial Intelligence Instructor

Dec 2024 - Aug 2025

Islamabad, Pakistan

- Designed and delivered the **Artificial Intelligence (Machine Learning & Deep Learning)** course at Corvit Networks, under the Prime Minister’s Special Initiative for Youth in collaboration with NAVTTC, training 20 students (Dec 2024 - Mar 2025).
- Delivered the **HCCDA-AI** (Huawei Cloud Certified Developer Associate - Artificial Intelligence) course at Corvit Networks, as part of Huawei’s mission to empower youth across Pakistan in collaboration with NAVTTC, training 50 students (May 2025 - Aug 2025).

PROJECTS

[1] Tasalli - AI Companion Chatbot

Aug 2026

Tools: TypeScript, React, Tailwind CSS, Supabase, Groq (Llama 3.3 70B)



- Built an AI companion chatbot that offers emotional support through natural, empathetic conversation.
- Submitted to the AI Factory native.builder Hackathon 2026.

[2] Talaash - Research Group Matching Platform

Mar 2026 - July 2026

Tools: React, Vite, FastAPI, Python, ChromaDB



- Built a platform that helps researchers find research groups that truly match their interests and goals.

[3] DIPCar: Autonomous Navigation over Economical Hardware

Mar 2024 - Oct 2024

Tools: Jetson Nano, SSD MobileNet, FCN ResNet34, TensorRT



- Designed and deployed a real-time autonomous navigation prototype on Jetson Nano, using SSD MobileNet for object detection and FCN ResNet34 for semantic segmentation, optimized with TensorRT.

Tools: CLIP model, Streamlit, PyTorch

- Implemented and evaluated deep learning models for facial emotion recognition using the FER-2013 dataset, achieving real-time performance with fine-tuned *openai/clip-vit-base-patch32*.

Tools: gemini-api, streamlit, youtube-transcript-api, python

- Developed a Streamlit application that extracts YouTube transcripts and generates concise notes using Google's Gemini API.

TECHNICAL SKILLS

Languages: Python, C/C++, HTML/CSS

Machine Learning & Deep Learning: PyTorch, Scikit-learn, Hugging Face, Ultralytics, TensorRT, NumPy, Pandas, Jetson-Inference

Computer Vision: OpenCV, CLIP, SSD MobileNet, FCN ResNet34, Roboflow

APIs & Deployment: FastAPI, Streamlit, Docker-free edge deployment (Jetson Nano), Vercel, Netlify

Tools & Platforms: Git, GitHub, Jupyter, Kaggle, Google Colab, Linux, Huawei ModelArts, Draw.io, LaTeX

Data Structures and Algorithms: 150+ LeetCode problems solved

PUBLICATIONS

- [1] NeuroShield-IoE: A Cognitive, Quantum-Resilient, and Self-Evolving Cybersecurity Framework for the Internet of Energy (Paper)

Oct 2025

IEEE Network
- Shehzad Ali*, Imran Nawar*, I. H. Lee, M. Sajjad, A. Alotaibi, V. H. C. de Albuquerque, K. Muhammad
- [2] Democratizing Autonomous Driving Research via Economical Hardware (Poster) (Code)

June 2026

Under Review at IJVAS
- Imran Nawar, Wajahat Ullah, Muhammad Sajjad

CERTIFICATIONS

- Computer Vision Basics

The State University of New York | Coursera - June 2025
- Neural Networks and Deep Learning

IBM | Coursera - Mar 2025
- Mathematics for Machine Learning: Linear Algebra

Imperial College London | Coursera - Jan 2025
- Getting Started with AI on Jetson Nano

NVIDIA - Sep 2024
- Machine Learning Specialization

DeepLearning.ai | Coursera - April 2024
- Fundamentals of Digital Image and Video Processing

Northwestern University | Coursera - Feb 2024
- Matrix Algebra for Engineers

The Hong Kong University of Science and Technology | Coursera - Dec 2023
- Python for Data Science, AI & Development

IBM | Coursera - Nov 2023
- Meta Front-End Developer Specialization

Meta | Coursera - Jul 2023
- Introduction to Python

Datacamp - Oct 2022

HONOURS AND AWARDS

- Recipient of a Project Based Stipend Scholarship, funded by Australia's Economic Accelerator Innovate Grant, to undertake a Master of Applied Research at Victoria University, Melbourne, Australia

2025
- Recipient of FYP Research Grant from Ignite NGIRI (PKR 88,644)

2024
- Recipient of Prime Minister's Laptop under the PM's Youth Laptop Scheme (Phase III)

2023